AUTHORIZATION			
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REVISION HISTORY				
REV. #	**CO #**	**DESCRIPTION**	**PREPARED BY**	**DATE**
05.01	NA	Moved some detail from the Quality Manual to this SOP and from this SOP to the new work instructions relating to it.	Alexey Sorokin, Scrum Master	Jan 24, 2011

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1. Purpose

The purpose of this procedure is to define the methods for identifying, compiling, reviewing and determining the requirements for the design input, which is the basis of the product specification, and to address requirements for compliance with all applicable standards, laws and regulations.

2. Scope

This procedure applies to all development projects, of products that are defined as medical device (not exempted from quality system) in the target market, in PeriGen, including new designs and changes made to existing products. Design and development projects performed for PeriGen by a subcontractor shall comply with PeriGen Design Control procedures and policies or with equivalent procedures.

3. Definitions and Abbreviations

Design Input: The physical and performance requirements of a device that are used as a basis for device design and development.

DHF - **Design History File**
SDLC - Software Development Life Cycle

4. Applicable Documents

Standards and Regulations:

ISO 13485:2016 sections 7.3.3
Health Canada Medical Devices Regulations, sections 10-11, 13-16, 18, 19, 32
FDA Quality System Regulations – 21CFR Part 820 section 30.

Procedures:

QMS-7.3.1 - Design and Development Planning.
QMS-7.3.3 - Design Output, Design Verification and Design Validation.
QMS-7.3.4 - Design Review.
QMS-7.3.7 - Design and Development Changes.
QMS-7.3.8 – Design Transfer and Product Release.

Work Instructions:

RA-WI-002– FDA Regulatory Requirements.
RA-WI-003 - Canadian Regulatory Requirements
RD-WI-001 – Risk Management.
RD-WI-002 – Design History File.

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SW-WI-001 – Software Development Life Cycle.

5. Procedure

5.1 Development of a new product or change in an existing product may be initiated internally or externally as follows:

5.1.1 *Internally:* company employee who recognizes a specific market need and/ or user input obtained from analyzing customer complaints and customer feedback, an opportunity to implement existing technologies, or a possibility to improve existing products.

5.1.2 *Externally:* customer or subcontractors who are interested in utilizing company technologies for improving an existing product or for developing a new product, which will be designed to answer a specific, need.

5.2 Following a feasibility study or other assessments the management will decide whether to implement the suggested design or design change as described in design planning procedure QMS-7.3.1 Design and Development Planning.

5.3 After customer/ marketing requirements are defined, a meeting will be held to discuss the project and its implications. The forum meeting shall include personnel from the various areas: R&D, SW Development, Installation, Marketing, Management, Finance, Clinical, QA, Regulatory, etc., as applicable to the specific product to be developed.

5.4 Following the meeting, the Development Project Manager will review the requirements for the product and their compliance with acceptable company parameters, and will detect parameters that need to be addressed. According to the specified requirements and review of relevant standards, the Development Project Manager will prepare the input document(s) that shall include all applicable requirements and issues that are derived from them.

5.5 As necessary, the Development Project Manager will approach company personnel, the project initiator, medical practitioner, subcontractors and/ or external consultants to clarify and resolve indecisive or contradicting requirements and to obtain input for undefined parameters. The requirements document shall be corrected and updated accordingly.

5.6 The requirements document shall include, as applicable to the specific project, the following details:

5.6.1 Introduction- description of the project, description of the clinical problem to be solved, overall description of the product, its intended use, intended user and objectives.

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5.6.2 Applicable Documents- list of documents that serve as the basis for the input document(s) such as standards, regulations, marketing/ user input document etc.

5.6.3 Product Requirements- such as:

- Functional requirements
- Human factors/ usability
- Performance requirements
- Configuration requirements
- Clinical support standards and guidelines
- Safety and reliability requirements
- SW and GUI Requirements (as the basis for the SRS document)
- Product classification and related regulatory requirements of Health Canada and the US as applicable to the planned target markets etc.)
- Compliance with mandatory standards and regulatory requirements at the target market, including standards such as QMS relating standards SW development related standards, FDA regulations and Guidance documents, etc.
- Voluntary standards
- External interfaces (e.g., connection to/ compatibility with other devices).
- Interoperability (e.g. connection to other networks)
- Security and Cybersecurity
- Documentation requirements, labeling (such as Operator's or User Manual)
- Installation requirements
- Service, support and user training requirements.
- Product lifetime
- Risk Analysis output (i.e. requirements that originate from risk mitigation activities) – each such input will be added to the various document chapters according to its relevance

5.7 The design input document(s) will be reviewed by professionals from the various areas, who are required to provide input, detect unresolved issues, to ensure that requirements are complete, unambiguous and non-conflicting, and are able to be verified or validated, following **QMS-7.3.4** *Design Review* procedure.

5.8 The design input document(s) will be approved by the VP Product / SVP R&D (Name, signature and date). After approval, the input document becomes a controlled document under the responsibility of the Development Project Manager.

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5.9 The requirement document shall be detailed to an engineering level of details and include a full set of requirements. The requirements shall be measurable and define a range or limits for the required attribute/ feature, rather than define the design and development solution/ parameter for this attribute/ feature.

5.10 The current requirement document shall be filed in the DHF under the Design Input section; old version shall be marked as "Obsolete" and kept in the DHF under the obsolete files section.

5.11 The Development Project Manager is responsible to update and distribute the requirement document at the different project phases. Each version shall be signed by the Development Project Manager and the VP Product /SVP R&D and bear a date and version number.

5.12 At advanced project stages the Development Project Manager will use a traceability matrix, or a similar tool, to assure that the project's design output and verification activities address all requirements and that all risk mitigations are addressed in the requirement document.

6. Responsibilities

6.1. The VP Product /SVP R&D has the overall responsibility for all development activities and for the implementation of this procedure.

6.2. The Development Project Manager is responsible to prepare or oversee the preparation of the design input document(s) and the traceability matrix, in accordance with this procedure.

6.3. All personnel involved in the project are responsible to provide the proper support for the preparation of the design input document(s).

6.4. QA Manager or delegate is responsible for managing and maintaining all Design History File records in compliance with PeriGen's **QMS-4.2.4** *Control of Quality Records* and **QMS-4.2.3** *Documentation Controls.*

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